[1] “Security vs privacy vs anonymity” pg 22. Turn Off Your Phone.org https://www.fightforthefuture.org/wp-content/uploads/2025/09/FullGuide-TurnOffYourPhone.pdf

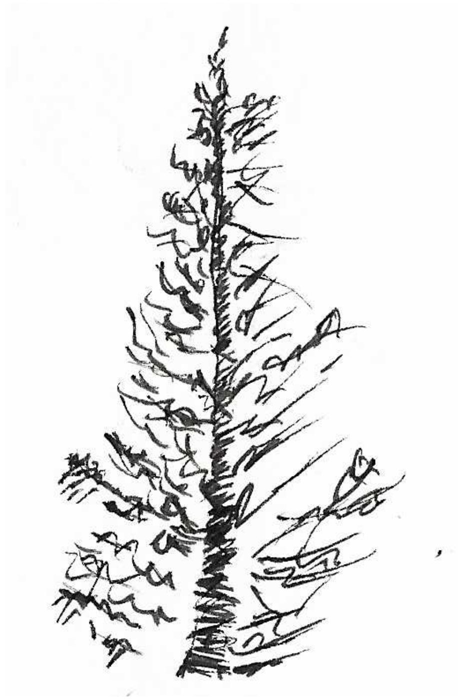
[2] “The Basics” Thread Modeling Fundamentals Håkan Geijer https://opsec.riotmedicine.net/downloads#threat-modeling-fundamentals

[3] https://tails.net/doc/first_steps/start/pc/index.en.html#boot-menu-key

[4] https://www.tomshardware.com/reviews/bios-keys-to-access-your-firmware,5732.html

PDX Community Tails OS Workshop

v0.2



10 Setting up a Password Manager with KeePassXC

11 Using MetaData Cleaner

12 Using OnionShare

Ways to share the OnionShare URL and or private key from within TailsOS.

Note: none of these options are end-to-end encrypted, but can be fairly anonymous.

- A throwaway or anonymous email address.
  - Tuta mail lets you create email addresses without an additional verification info. You just have to wait for a few days to complete verification.
  - Gurrillamail.com lets you create disposable email addresses.
- Riseup pad
  - You still have to send the URL/private key for the Riseup pad, but it can be pre-shared either in person or via a different means.
- Browser based chat

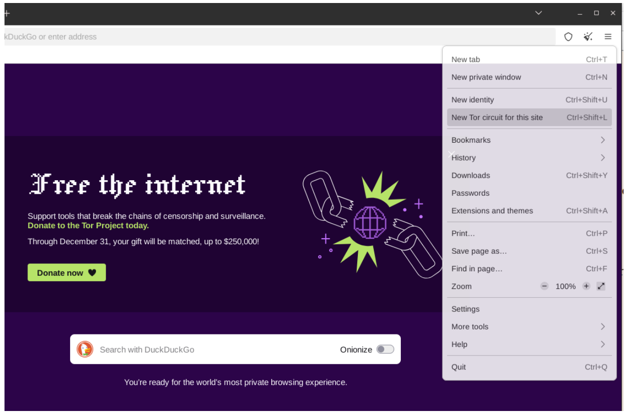
The following methods require either installing additional software into your Tails Usb, or copying files to an extra usb to transferring to another computer that has this software, both of which present different kinds of security compromises.

- Signal (or other messenger app like Discord, ect.)
- Magic Wormhole
- Bitwarden Sendx'
- Warpinator (Winpinator for Windows)
- Ect.

13 Creating a LUKS Encrypted Drive

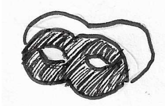
References

[1] "Motor Vehicle Traffic Crashes as a Leading Cause of Death in the United States, 2016, 2017", NHTSA. https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/812927



9.3 Preserving Anonymity

Tails can't preserve your anonymity online if you supply information that identifies you. If you are trying to remain anonymous during your session, don't log into any accounts that you use on other devices. If you create an account inside of a TOR session, don't log into that account outside of TOR.



9.4 Onion Links

Onion URLs are web services that are served over TOR, and only accessible via TOR. Here's an example, the onion address for Duck-Duck-Go: https://duckduckgogg42xjoc72x3sjasowoarfbgcr

Here is a Github repo of onion sites:
https://github.com/alecmuffett/real-world-onion-sites?tab=readme-ov-file#education

Part I
Intro

1 Intention

Technology rewards the curious.

I am running this workshop to equip my community with tools to better protect its safety.

That said, it can be very easy to become overwhelmed and fatalistic, obsess over worst case scenarios, or become fixated on mythical "most secure" or "correct" tools and processes. (The only way to be perfectly secure is to talk to no one, do nothing of consequence, and never take a single risk).

We learn best when we are, curious, playful, and consistently engaged. While reading a list of expert recommendations once probably makes you a little safer than nothing, you are far better served improving your safety skills a little bit every week. Trying new tools, researching, and discussing it with people, especially people you don't necessarily agree with. If you find learning about computers or online privacy miserable, you aren't going to do any of that.

I will personally consider the workshop a success if everyone who attends learns:

- Something they think is cool.
- Something that makes them curious to know more.
- Something that equips them to be one modicum safer than they were before.

2 What Is Tails?

Tails (The Amnesic & Incognito Live System) is a tool for mitigating digital surveillance and censorship.

It is a bootable Live USB. You plug it into a computer, and boot into a temporary separate Operating System.

By default Tails will:

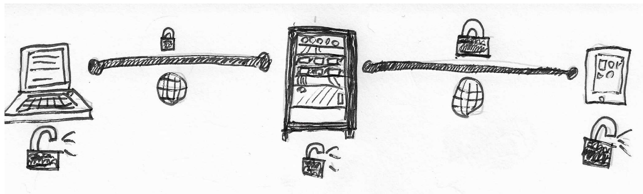
- Retain no data between sessions to avoid leaving traces of what you use it for.
- Route all internet traffic over the TOR network to provide privacy and anonymity while using the internet.

3 Terminology

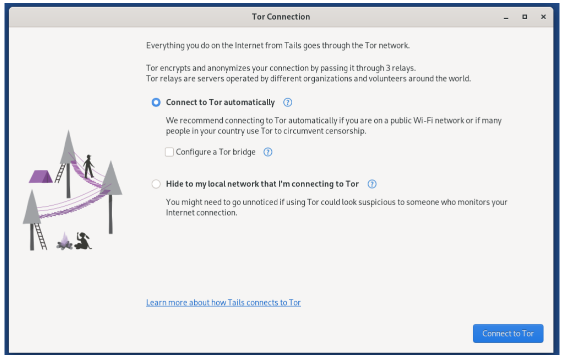
Some very quick definitions. A shared understanding of these terms makes us more effective when talking about digital privacy.

- Anonymity - Who is talking to who? Metadata.
- Privacy - What are they talking about?
- Security - How resistant is their communication medium to interventions that downgrade it in potentially several ways[1]:
  - Anonymity, unmask the participants.
  - Privacy, gain access to the contents of their communication.
  - Availability, take the medium down and make it unavailable for use.
  - Otherwise co-opt or use the medium for unintended purposes.

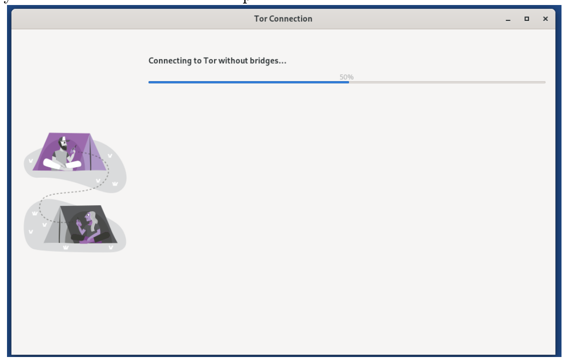
- Encrypted in Transit
  - Most websites use a technology called SSL, using the "HTTPS" prefix. This means the traffic is encrypted between the client and the server, and can't be read by the ISP or anyone else listening on the network. It can still be read by the server.
  - For example, Discord is a messaging app that is encrypted in transit.



- End-To-End Encrypted (E2EE)



Hit "Connect" and the system will begin setting up your TOR circuit. It may take several minutes to complete.



9.2 Sites that Block TOR

Some sites block access from the TOR network. One thing you can try is to use a new TOR circuit. Not all TOR exit nodes may be blocked by the site you are trying to visit. To request a new TOR circuit, select the options menu in the top right hand corner of the TOR browser, and select "New Tor Circuit for this Site".

Note, if you are trying to start a new session to preserve anonymity, it is safer to restart TOR completely or restart Tails.

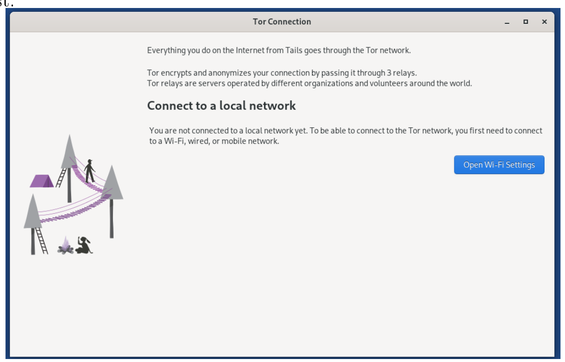
Part III
Using Tails

Tails has a handful of tools for doing basic computing tasks, the TOR browser for web surfing, a basic Office suite, an offline password manager, sound and video players, and some system tools.

9 Using TOR

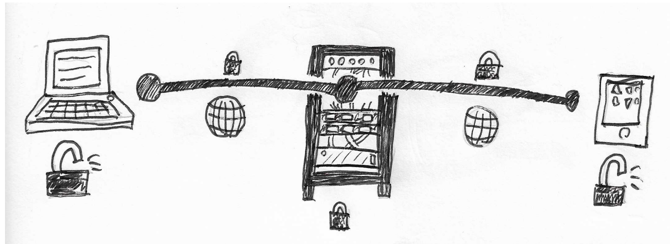
9.1 Connecting to TOR

When you first launch TOR, you will be presented with a screen for configuring your TOR session. If you are not connected to the Wifi or a network, do that first.

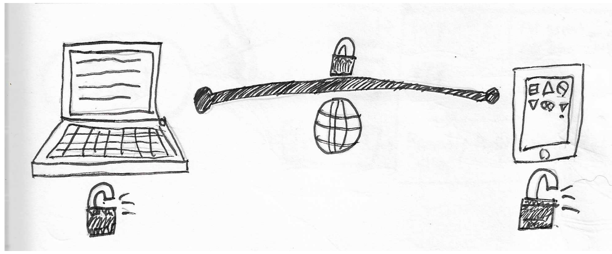


In the following menu, select "Connect to TOR Automatically". You will be given the option to connect with a TOR bridge. TOR bridges are essentially a VPN for the first hop of the TOR circuit, providing a way to access the TOR network in countries that block it, or if you want to hide from your ISP you are using TOR. This may or may not be necessary given your threat model and risk tolerance.

- Services that are end-to-end encrypted mean the data is encrypted in transit, and also encrypted on the server. This is usually used to refer to messaging apps where only the intended recipient can decrypt the message. Typically a server is involved, routing the encrypted data from one client to another. End-to-end encrypted servcies offer privacy for the contents of messages, but dont usually offer anonymity, as metadata like who is talking to who, and when, is still visible to the server.
- Signal is end-to-end encrypted, but not anonymous, and requires users to provide a phone number to use the service.



- Peer-To-Peer (P2P)
  - Services that are peer-to-peer don't involve a central server at all. Instead individual clients connect to directly to one another. Usually this means both parties must be online at the same time, and have a way of pairing their devices either in person at a previous point, or via a different communication method.
  - Briar is a peer-to-peer messenger app.
  - SimpleX and Delta Chat are mostly peer-to-peer, relying on servers to connect peers but take steps to run multiple servers by different parties and minimize the amount of metadata any individual server has.



- Difficulty Differential
  - Mitigation's that are easy for me, but hard for an attacker are the most valuable. Some mitigation's will be more like 50/50, these are less useful but not useless depending on your needs.
- Risk

  Risk = Cost of bad thing * Likelihood of bad thing

  Examples:

  Risk of getting in a car crash:

  Cost: Pretty high, ranges from injury or death depending on the kind of accident.

  x

  Likelihood: Depends on your age, lifestyle, and other factors. In the United States it ranked as the 13th leading cause of death among all causes in 2016 and 2017.[1]

  = Risk depends on individual factors.
- TOR
  - TOR stands for "The Onion Network" and is a privacy preserving network. The TOR network is typically accessed with the TOR browser. TOR routes all traffic through at least 3 encrypted relay nodes before being routed to its final destination. Traffic is encrypted in layers, with each relay node removing a single layer of encryption. This is designed to prevent any one node from knowing the source and destination of your web traffic.
- Onion Services
  - Onion services are websites that can only be reached via the TOR network. Some websites are only accessible via Onion services (the so-called "Dark web"), while other websites have clearnet sites, while also offering Onion sites as a privacy preserving option for users.

7.3 Changing the Boot Order in the BIOS/UEFI menu

1. If the other approaches fail, you can access your computer BIOS or UEFI and change the order of boot devices.
2. To bring up your BIOS/UEFI, a specific button needs to be pressed when the machine starts. Below is a table of the common buttons for bringing up the BIOS/UEFI. [4]
   (a) ASRock: F2 or DEL
   (b) ASUS: F2 for all PCs, F2 or DEL for Motherboards
   (c) Acer: F2 or DEL
   (d) Dell: F2 or F12
   (e) ECS: DEL
   (f) Gigabyte / Aorus: F2 or DEL
   (g) HP: F10
   (h) Lenovo (Consumer Laptops): F2 or Fn + F2
   (i) Lenovo (Desktops): F1
   (j) Lenovo (ThinkPads): Enter then F1.
   (k) MSI: DEL for motherboards and PCs
   (l) Microsoft Surface Tablets: Press and hold volume up button.
   (m) Origin PC: F2
   (n) Samsung: F2
   (o) Toshiba: F2
   (p) Zotac: DEL
3. Power on your computer and hit the button over and over, this should bring up the BIOS/UEFI for your machine. You may have to experiment with different buttons to find the right one.
4. Once in your BIOS/UEFI, there should be an option to manipulate the order of default boot devices. You will want to make USB the highest priority boot device.

8 Selecting Boot-up Options

Create encrypted partition
Mention enabling root password for installing additional software.

Manufacturer	Key
Acer	F12, F9, F2, Esc
Apple	Option
Asus	Esc
Clevo	F7
Dell	F12
Fujitsu	F12, Esc
HP	F9
Huawei	F12
Intel	F10
Lenovo	F12, Novo
MSI	F11
Samsung	Esc, F12, F2
Sony	F11, Esc, F10
Toshiba	F12
Others…	F12, Esc

[3]

4 Why Should I Use Tails? Things Tails Can and Cannot do.

4.1 What Tails Does well:

- Keeps web traffic anonymous and private from internet service providers or other people on the network.
- Keeps you anonymous from the website you are visiting, provided you don't supply identifying information.
- Provides decent security against most threats.

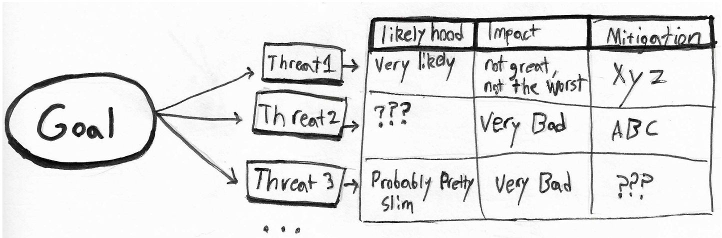
4.2 What Tails Doesn't do well:

- Hide from your Internet Service provider you are using TOR.
- Hide from the website you are visiting that you are using TOR.
- Protect your anonymity if you do something that unmasks you.
- Provide security against threat actors with nation-state level resources.

4.3 Threat Model

Whether Tails is an appropriate tool for you depends on what you are trying to do and your threat model.

- A Threat Model is a structured way of thinking about your goals and obstacles or people who may not want to you accomplish those goals. Several formalized frameworks exist. I'll be using the framework outlined in the Zine "Threat Modeling Fundamentals" by Håkan Geijer. [2]
  - What are our goals?
    - Who or what might oppose or complicate those goals? What kinds of tools do they have? What risk do they pose? What would the impact be?
    - What can we do about it? Do we have a mitigation that brings risk down to negligible levels? What about mitigations that reduces risk a little, but still has a tangible likelihood? Or is this a risk we are forced to accept?
    - Is our threat model well informed? Is it effective?



- Below are additional resources on threat modeling
  - Frameworks:
    * HARMS framework focuses less on technical treats and more on human impact: https://arxiv.org/html/2502.07116v1
    * This zine explores threat modeling in more detail and is tailored towards activists: https://opsec.riotmedicine.net/downloads#threat-modeling-fundamentals
    * One potential place to research threats and mitigations: https://www.notrace.how/threat-library/

Part II
Creating and Setting up Tails

5 Downloading the Tails ISO

- Navigate to Tails.net and download the Tails ISO.
  - https://tails.net/install/download/index.en.html
- Verify the download using the verification tool on the download page.

6 Flashing the Tails USB

- For Windows use Rufus
  - https://tails.net/install/windows/index.en.html
  - https://rufus.ie/en/
- For Mac use Raspberry Pi Imager
  - https://tails.net/install/mac/index.en.html

  - https://www.raspberrypi.com/software/
- For Linux you can use GNOME Disks or ‘dd‘
  - https://tails.net/install/linux/index.en.html

7 Booting From the USB

There are a few ways to boot from the USB, depending on the kind of laptop you have.

7.1 Booting from within Windows

If you are using Windows 8 or 10 you can start the process of booting to Tails from within Windows. https://tails.net/doc/first_steps/start/pc/index.en.html#windows-restart

1. Make sure the Tails USB is plugged in.
2. Click Start
3. Hold “Shift” while selecting “Power->Restart”
4. The machine will restart, but will boot to a menu of diagnostic options. Select “Use a device”.
5. If your Tails USB stick comes up in the list of options, select it.
6. Otherwise, if “Boot Menu” is an option, select that.
7. If neither of those options are present, you will have to try launching Tails using the Boot Menu Key.

7.2 Booting with the Boot Menu Key

1. Plug in the Tails USB.
2. Shut down the computer.
3. Computers from various manufacturers use different buttons to enable the user to bring up a boot menu when the computer starts up. Refer to the table below for common buttons associated with different manufacturers. You may have to experiment with different buttons to find the right one for your computer.
4. Start the computer, repeatedly mashing the boot menu key for your computer. If successful, a menu of boot options will come up. You may fail the first few attempts if you miss the window or if the boot menu key is an unexpected key.